

Network Intrusion Detection System

Problem Statement and Literature Survey

1. Problem Statement

Network intrusion is a major cybersecurity threat in which attackers attempt to gain unauthorized access, disrupt services, steal information or misuse network resources. Traditional signature and rule-based intrusion detection approaches are effective for known attack patterns but can struggle with new, modified or previously unseen attacks. Machine-learning and deep-learning techniques can learn patterns from network traffic and automate the identification and classification of malicious activity.

Nevertheless, existing AI-based Network Intrusion Detection Systems (NIDS) face important challenges including dependence on benchmark datasets, class imbalance, high false alarms, changing attack behaviour, computational cost, privacy constraints and weak generalisation to unseen network environments. Recent research therefore explores deep learning, explainable AI, federated learning, adversarial robustness and more realistic evaluation methods.

Therefore, the proposed project aims to develop a Network Intrusion Detection System that collects or uses network-flow features, preprocesses the data, trains and compares suitable machine-learning models, evaluates them using accuracy, precision, recall and F1-score, and provides a clear normal/attack decision. The system should emphasise practical detection, low false alarms and the ability to generalise to realistic traffic.

2. Literature Survey

1. NIDS-CNNLSTM: Network Intrusion Detection Classification Model Based on Deep Learning

Author(s): Authors of the IEEE Access study (2023)

Algorithm / Methodology: Combines Convolutional Neural Network (CNN) feature learning with Long Short-Term Memory (LSTM) sequence modelling. It evaluates binary and multi-class intrusion detection using KDD CUP99, NSL-KDD and UNSW-NB15 datasets.

Advantages: Combines spatial/feature extraction with sequence learning; supports binary and multi-class detection; evaluated on multiple classic datasets; reports improved detection performance and low false-alarm behaviour compared with earlier models.

Disadvantages / Limitations: CNN-LSTM architectures are more computationally demanding than simple ML models. The evaluation relies on benchmark datasets, so performance on current enterprise traffic and unseen attacks may differ.

2. An Explainable Ensemble Deep Learning Approach for Intrusion Detection in Industrial Internet of Things

Author(s): Authors of the IEEE Access study (2023)

Algorithm / Methodology: Uses an ensemble deep-learning IDS for IIoT networks and incorporates SHAP and LIME to explain model decisions. The ToN_IoT dataset is used for evaluation.

Advantages: Improves transparency by explaining predictions; ensemble learning can improve detection performance; provides information useful to security experts who need to understand why traffic is considered malicious.

Disadvantages / Limitations: Explainability methods add processing and implementation complexity. The study focuses on IIoT and a specific dataset, so broader enterprise-network generalisation requires additional validation.

3. A Comprehensive Survey of Generative Adversarial Networks (GANs) in Cybersecurity Intrusion Detection

Author(s): IEEE Access study (2023)

Algorithm / Methodology: Reviews the use of GANs in intrusion detection, including generation of synthetic attack samples, adversarial examples and augmentation of rare attack classes. It discusses datasets, evaluation metrics and cybersecurity use cases.

Advantages: Addresses class imbalance and scarcity of rare attack samples; can support adversarial-robustness research; provides a broad view of how generative models can be used in IDS research.

Disadvantages / Limitations: Generated samples may not perfectly represent real attacks. GAN-based pipelines can be difficult to train and add computational complexity, so they may not be ideal for a lightweight first-stage NIDS.

4. An Efficient Federated Learning System for Network Intrusion Detection

Author(s): IEEE Systems Journal study (2023)

Algorithm / Methodology: Proposes Dynamic Weighted Aggregation Federated Learning (DAFL), where multiple participants train local IDS models while sharing model information rather than raw security data. Dynamic filtering and weighting are used to reduce communication overhead.

Advantages: Helps preserve data privacy; allows organisations to collaborate without directly sharing raw network data; designed to reduce communication overhead while maintaining detection performance.

Disadvantages / Limitations: Federated learning introduces coordination, communication and model-aggregation complexity. Non-IID data and differences between organisations can still make training and generalisation difficult.

5. Machine Learning on Public Intrusion Datasets: Academic Hype or Concrete Advances in NIDS?

Author(s): IEEE/IFIP DSN-S study (2023)

Algorithm / Methodology: Critically examines the widespread use of public intrusion datasets in ML-based NIDS research and questions whether high benchmark performance reliably represents real-world progress.

Advantages: Highlights the importance of realistic evaluation; draws attention to dataset construction, experimental methodology and the gap between benchmark results and operational networks.

Disadvantages / Limitations: It is primarily a critical methodological study rather than a single deployable detection algorithm. It does not by itself provide a complete NIDS implementation.

3. Comparative Literature Survey

Paper	Algorithm / Methodology	Advantages	Disadvantages / Limitations
NIDS-CNNLSTM	CNN + LSTM; binary and multi-class detection; KDD CUP99, NSL-KDD, UNSW-NB15.	Learns complex and sequential traffic patterns; supports multiple attack classes.	Higher computational cost; benchmark-dataset dependence.
Explainable Ensemble DL for IIoT	Ensemble deep learning with SHAP and LIME; ToN_IoT.	Improves transparency and helps analysts understand predictions.	Added explainability complexity; IIoT-specific evaluation.
GANs in Cybersecurity IDS	GAN-based sample generation, augmentation and adversarial examples.	Useful for rare classes and robustness research.	Synthetic data may differ from real attacks; computationally complex.
Efficient Federated Learning NIDS	DAFL with local training, dynamic filtering and weighted aggregation.	Privacy-preserving collaboration and lower data-sharing requirements.	Communication, aggregation and non-IID-data challenges.
Public Intrusion Datasets Study	Critical analysis of ML/DL experiments on public intrusion datasets.	Emphasises realistic evaluation and dataset quality.	Not a standalone detection algorithm; focuses on research methodology.

4. Research Gap Identified

- **Dataset dependency:** High accuracy on benchmark datasets may not transfer directly to real enterprise networks.
- **Generalisation:** Attack techniques continuously change, making previously unseen attacks difficult to detect reliably.
- **False positives:** A practical NIDS must reduce unnecessary alerts so that security teams can focus on genuine threats.
- **Class imbalance:** Some attack categories are much rarer than normal traffic or common attacks, affecting model training.
- **Real-time deployment:** Complex deep-learning models can increase processing and infrastructure requirements.
- **Explainability:** Security analysts need understandable reasons for why traffic has been classified as suspicious.
- **Privacy and collaboration:** Organisations may not be able to share raw network traffic, motivating privacy-preserving approaches such as federated learning.
- **Practical evaluation:** NIDS research should include realistic validation and, where possible, testing across different datasets or environments.

5. References

1. NIDS-CNNLSTM: Network Intrusion Detection Classification Model Based on Deep Learning, IEEE Access, Vol. 11, 2023. DOI: 10.1109/ACCESS.2023.3254915.
2. An Explainable Ensemble Deep Learning Approach for Intrusion Detection in Industrial Internet of Things, IEEE Access, Vol. 11, 2023. DOI: 10.1109/ACCESS.2023.3323573.
3. A Comprehensive Survey of Generative Adversarial Networks (GANs) in Cybersecurity Intrusion Detection, IEEE Access, Vol. 11, 2023. DOI: 10.1109/ACCESS.2023.3296707.
4. An Efficient Federated Learning System for Network Intrusion Detection, IEEE Systems Journal, Vol. 17, Issue 2, 2023. DOI: 10.1109/JSYST.2023.3236995.
5. Machine Learning on Public Intrusion Datasets: Academic Hype or Concrete Advances in NIDS?, IEEE/IFIP DSN-S, 2023. DOI: 10.1109/DSN-S58398.2023.00038.

Web sources used for verification: IEEE Xplore records for the selected NIDS studies. The literature survey distinguishes reported methodology/results from limitations inferred from each study's scope and evaluation design.